

AI-Driven Digital Twin for Millimeter-Wave Photonic Fronthaul Links

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Abstract—This paper introduces a machine learning-based Digital Twin for Optical Networks (DTON) to model and optimize the performance of millimeter-wave (mmW) photonic fronthaul links. Using an experimentally validated simulation framework, we generate a dataset of 1,100 configurations linking key system parameters to multiple Quality of Transmission (QoT) metrics. Three machine learning models—Gaussian Process Regression (GPR), Artificial Neural Networks (ANNs), and Extreme Gradient Boosting (XGBoost)—are trained to predict these metrics, with ANN and XGBoost demonstrating superior accuracy. Furthermore, Particle Swarm Optimization (PSO) and Genetic Algorithms (GA) are applied to identify optimal system configurations that enhance performance while minimizing power consumption. The results highlight the potential of AI-driven digital twins for real-time network optimization, paving the way for more adaptive and efficient next-generation optical-wireless systems.

Index Terms—Microwave photonics, machine learning, cloud radio access network, digital twin.

I. INTRODUCTION

Global mobile data traffic is expected to surge, potentially reaching 5016 exabytes per month by 2030, driven by disruptive technologies such as immersive and holographic communication, multisensory internet, and autonomous vehicles. The development of 6G networks aims to support this increase, potentially connecting around 97 billion mobile and machine-to-machine (M2M) devices by 2030 [1].

Future mobile network architectures are expected to embrace dense networks with increased data traffic and smaller cells, in a centralized-radio access network (C-RAN) model [2]. Integrating wireless and optical technologies within C-RAN can help to support high throughput, minimal latency, broad coverage, and accurate positioning, with baseband units (BBUs) located at the central office (CO) and linked to remote radio units (RRUs) through fronthaul (FH) connections.

Radio-over-fiber (RoF) technology offers solutions for low-latency, high-speed data transmission. Microwave-photonics techniques enhance performance through low phase noise, frequency tunability, and cost-effective implementation, such as optical frequency multiplication via an external Mach-Zehnder modulator (MZM), which generates high-quality signals at the millimeter-wave band. Analytical and experimental studies

demonstrate that using a directly modulated laser (DML) and a dispersive fiber link leads to a substantial electrical gain in millimeter-wave (mmW) signal generation [3], which is especially promising under small signal regime towards energy-efficient network solutions suitable for future deployments.

As demand for mobile networks continues to grow, advanced tools to manage the complexity of next-generation networks are also needed. Digital twins for optical networks (DTONs) offer a promising solution in this area. A digital twin—a virtual model of a physical network—enables real-time monitoring, predictive analysis, and fast reconfiguration of network links. Compared to other existing tools based on telemetry, like Optical Performance Monitoring (OPM), that use hardware sensors to measure key optical parameters and face high deployment costs and limited scalability, the DTON approach helps to optimize the performance and resource usage, as well as to improve the efficiency in densely connected network nodes [4].

However, achieving a reliable DTON capable of real-time operations presents several challenges, such as variability in physical hardware, incomplete system observability, and the dynamic nature of optical environments that complicates the task of creating robust, generalizable models. Moreover, even with high-accuracy DTONs, additional techniques are required to continually refine configuration parameters and optimize performance amongst fluctuating network demands.

To create accurate digital twins, approaches range from modeling individual components to simulating entire systems. These methods fall into two categories: analytical models based on mathematical formulations of physical parameters and data-driven approaches that use machine learning (ML) to identify patterns from observed data. Often, hybrid models combine both approaches to enhance predictive accuracy. An extensive review of these state-of-the-art approaches—referred to as mirror modeling—is presented in [4]. Several studies focus on directly predicting Quality of Transmission (QoT) metrics such as Signal-to-Noise Ratio (SNR), Generalized SNR (GSNR), and Q-factor by capturing the impact of various physical and environmental factors on signal quality, which is the approach we adopt.

One common analytical approach is the Gaussian Noise (GN) model, which approximates signals as stationary Gaussian noise and treats nonlinear interference as an additive term,

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assuming that nonlinear effects are sufficiently small for a perturbative approximation [5]. However, the GN model may be inaccurate for low-dispersion or short-fiber systems. An enhanced GN (EGN) model was proposed to address these limitations [6], albeit with increased complexity. A limitation of analytical models is that they depend on detailed physical parameters, such as fiber dispersion coefficients, fiber loss, noise figures of erbium-doped fiber amplifiers (EDFAs), and nonlinear coefficients.

Fully data-driven ML models, typically optimized via supervised learning, perform well in trained setups but may struggle with unseen configurations [7]–[10]. Techniques such as transfer learning [9] and setup-agnostic architectures like invariant convolutional neural networks [10] attempt to mitigate this issue. Hybrid approaches that combine prior domain knowledge with ML are also attractive. For example, physics-informed models incorporate physical insights into the training process, reducing the required data volume and improving model generalization by grounding predictions in physical principles [11].

In this context, common architectures include artificial neural networks (ANNs) [8], [9] and Gaussian Process Regression (GPR) models [7], [11]. While ANNs may offer greater modeling capabilities, GPR models provide both an output estimation and an uncertainty measurement, making them appealing for real-world applications. However, both approaches have notable limitations. ANNs require large datasets to optimize their parameters effectively, while GPR models become computationally expensive with large datasets, as kernel optimization involves a matrix inversion that scales poorly with the number of samples ($O(n^3)$) [11].

The main goal of this paper is to develop a ML-based digital twin for predicting QoT metrics in a mmW photonic fronthaul link. This work aims to enhance network performance and resource optimization by leveraging artificial intelligence to model and optimize signal transmission in optical fronthaul networks, which are critical for 6G and future mobile networks. Specifically, we evaluate the performance of three ML techniques for this task: GPR, ANNs, and extreme gradient boosting (XGBoost) [12]. We aim to quantify the limitations of the two state-of-the-art models, introduce an additional promising model into the discussion, and determine the most suitable algorithm for this task. We also compare Particle Swarm Optimization (PSO) [13] and Differential Evolution (DE) [14]—a genetic algorithm (GA)—for resource usage optimization.

In contrast to most QoT models that predict a single figure of merit (FoM), our approach aims to predict five different quality metrics—received optical power (RoP), electrical power (ReP), error vector magnitude (EVM), signal-to-noise-ratio (SNR) and modulation error ratio (MER)—making it uniquely suited to provide a more comprehensive assessment of system performance. To generate the data employed for training the models, we utilize a simulation setup validated by experimental data. We vary three input parameters that are fully configurable in the optical setup, so our approach does

not require prior knowledge of the network’s physical parameters, reducing dependency on potentially noisy or uncertain physical measurements. Although this may limit the model’s generalizability, this limitation can potentially be overcome through transfer learning, as demonstrated in [9].

II. PHOTONIC MMW SIGNAL GENERATION SETUP

As part of the proposed framework, Fig. 1(a) illustrates the complete simulation setup, encompassing the photonic mmW signal generation system within a C-RAN architecture. At the baseband unit located at the CO, a DML (Emcore 5021-T) emitting an optical carrier at $f_0 = 193.1$ THz with 8.5 mW optical power and a 3 dB bandwidth of 5 GHz is modulated by a QPSK data signal of variable bandwidth [100, 1000] MHz and an electrical power P_{data} in the [-30, 15] dBm range at an intermediate frequency, $f_{\text{IF}} = 2$ GHz, which is generated by an arbitrary waveform generator (AWG) (Tektronix AWG7122C). The state of polarization is adjusted by a polarization controller (PC) and the optical signal is launched into the MZM (Sumitomo T.DEH1.5-40X-ADC-Y-Z), which is biased at the null transmission point (V_π), to obtain optical carrier suppression. The MZM is driven by a radiofrequency (RF) 20 GHz signal tone with an electrical power, $P_{\text{RF}} = 23$ dBm, generated by a signal generator (SG) (Agilent 8267C). The optical signal is amplified by an EDFA (Amonics AEDFA-23-B-FA) to compensate the optical losses with 13 dBm fixed optical output power and, after that, the amplified spontaneous emission (ASE) noise is filtered out by an optical band pass filter (OBPF) (Alnair BVF-100) with a bandwidth of 2 nm. Then, the optical signal is transmitted over the standard single mode fiber (SSMF) link as the optical distribution network (ODN). More details of the experimental setup are provided in [3].

At the optical network unit (ONU), the optical signal power is set in the variable optical attenuator (VOA) and the optoelectronic conversion is held in the photodetector (PD) (Finisar BPDV2020R). Then, the mmW signal ($f_{\text{mmW}} = 2f_{\text{RF}} + f_{\text{IF}}$) is generated by the beating of the optical sidebands $f_0 \pm f_{\text{RF}}$ and $f_0 \pm f_{\text{RF}} \pm f_{\text{IF}}$ and thus, the quality of the transmitted data in the mmW band is analysed by a vectorial signal analyser (VSA) (Tektronix DPO72004C). As it was mentioned before, system performance has been evaluated by a set of FoM, i.e., the RoP, the ReP, the EVM, the SNR and the MER.

The experimental setup has been exhaustively characterized in a real optical link with typical inherent noise and distortions [3], and a simulation model has been developed using Optisystem software which faithfully reproduces the response of the real system described in Fig. 1(a). More concretely, a 20 MHz bandwidth QPSK signal has been transmitted with $f_{\text{IF}} = 2$ GHz and $P_{\text{data}} = -11.5$ dBm, and EVM measurements have been done vs RoP, ReP and MER, as depicted in Figs. 1(b.1), (b.2) and (b.3), respectively. Fig. 1(b) shows a good agreement between experimental results (symbols) and simulation curves (solid lines), that validates our simulation model to replicate the real system performance, and thus, validates it to generate the training data for the ML based

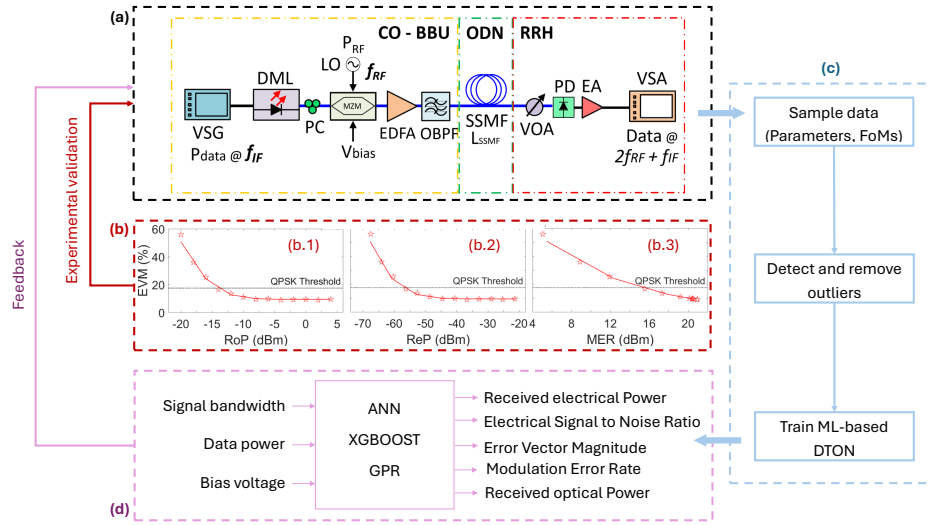


Fig. 1: Experimental/simulation setup of a photonic mmW signal generation in a C-RAN with the detail of workflow (c) and digital twin model block description (d). Insets depict experimental (a) and simulation (b) EVM results corresponding to a 20 MHz QPSK signal transmitted over 42 GHz vs: (b.1) RoP, (b.2) ReP and (b.3) MER. CO-BBU: baseband unit at the central office, VSG: vector signal generator, DML: directly modulated laser, PC: polarization controller, MZM: Mach-Zehnder modulator, LO: local oscillator, EDFA: erbium-doped-fiber amplifier, OBPF: optical band pass filter, ODN: optical distribution network, SSMF: standard single mode fiber, VOA: variable optical attenuator, PD: photodetector, RRU: remote radio unit, EA: electrical amplifier, VSA: vector signal analyzer.

DTON model. This approach is advantageous for generating large amounts of data, since using the simulator simplifies the database creation, avoids experimental issues, saves resources and reduces waiting times.

The proposed workflow for this system, outlined in Fig. 1(c), involves several stages: first, the experimentally validated simulation model provides the dataset, generating vectors that relate the input parameters and their respective output metric values. These data are then processed and curated to eliminate outliers, ensuring quality for subsequent analysis. When a sufficient amount of clean data is available, it is used to train the ML model, shown in Fig. 1(d). Once the model becomes reliable, it serves as a surrogate for the actual system, enabling real-time optimization of resource allocation to achieve target performance goals.

III. DATA GENERATION AND EXPLORATION

Utilizing the setup described in the previous section, the dataset has been generated by varying the power of the electrical signal (P_{data}), its bandwidth (BW), and the biasing voltage (V_{Bias}) of the MZM modulator. These parameters are directly configured in the Baseband Unit (BBU), allowing the system to operate independently of specific physical characteristics of the rest of components.

To evaluate the model's performance across various configurations, a number of equidistant values has been tested for each parameter within a given range, generating a total of 1100 unique combinations, as summarized in Table I. Note that each sample was generated under typical noise and distortion conditions to reflect the variability and uncertainty inherent in real-world scenarios.

TABLE I: Ranges of training parameter values for the system.

Parameter	Range	# Values
P_{data}	[-30, 15] dBm	10
BW	[100, 1000] MHz	10
V_{Bias}	[4.1, 6.1] V	11

An exploratory data analysis was conducted with three main goals: to assess the complexity of the sample space and determine the appropriate model size and training effort; to evaluate input relevance and identify potential information redundancy that could affect training efficiency; and to detect outliers that may arise from simulation anomalies and impact optimization performance.

Fig. 2(a) depicts the RoP values as dependent on V_{Bias} and it is possible to observe that every BW and P_{data} configuration lead to completely overlapped curves, resulting in a single line representation. This occurs because this metric depends solely on the V_{Bias} , so the output value is fixed regardless of the BW and P_{data} values. In fact, this curve corresponds to the power-voltage characteristic curve of the MZM.

Figs. 2(b) and (c) show EVM and MER, respectively, as functions of P_{data} . While these metrics exhibit minor variations with V_{Bias} , they change substantially with P_{data} and BW. Some unexpected outliers are observed that may hinder model accuracy, so omitting them could improve performance. Their removal produces smoother curves relative to P_{data} and BW, enabling potentially more accurate predictions. This hypothesis was confirmed by rerunning the model, which revealed that the outliers were specific errors in the prior simulation process.

Figs. 2(d) and (e) display ReP and SNR, respectively, also as functions of P_{data} . Similar to EVM and MER, ReP and SNR

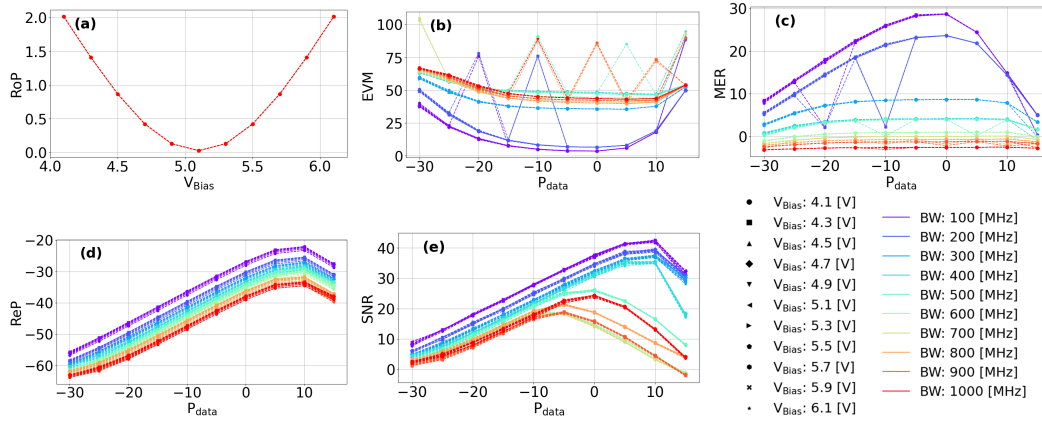


Fig. 2: Plot of FoM metrics obtained from the dataset generated with the setup depicted in Figure 1(a). (a), RoP vs V_{Bias} for different P_{data} and BW, (b) EVM vs P_{data} for different V_{Bias} and BW, (c) MER vs P_{data} for different V_{Bias} and BW, (d) ReP vs P_{data} for different V_{Bias} and BW, (e) SNR vs P_{data} for different V_{Bias} and BW.

show minimal variation with V_{Bias} but significant sensitivity to P_{data} and BW. However, no outliers are detected in these metrics, resulting in smooth curves, and thus no additional preprocessing is required.

Note the importance of the analysis conducted in this section prior to the development of the ML models, as it provides valuable insight into the interpretation and relevance of the various input parameters in relation to the QoT metrics.

IV. METHODOLOGY

Based on the previous analysis, we can draw two main conclusions regarding the implementation of the ML models: (1) To predict the RoP, only the V_{Bias} parameter is needed, as P_{data} and BW have no impact on this FoM and (2) Complex networks are unnecessary, given the relatively smooth relationships between the metrics and the parameters.

In this case, all parameters have been used as model input, and all metrics are predicted at the model output. If the metrics are correlated to some extent, it is often a good practice to build a single model with a multi-output goal, since the relations extracted during the training could be leveraged by various outputs as shared knowledge.

For performance comparison, GPR, ANN and XGBoost models are employed. To train and evaluate each model's performance, root mean squared error (RMSE) is used as the primary deviation metric for each FoM. To determine the amount of data needed for a target accuracy, each model is trained on nine independent data splits, incrementally increasing the size of the training set from 10% to 90% in steps of 10%, with the remainder used for testing. Within each training set, 80% of the data is allocated to model training and network optimization, while the remaining 20% serves as a validation set. This approach allows us to systematically evaluate model accuracy across varying data volumes, highlighting the trade-off between data availability and model performance—an especially critical factor given that data collection is often expensive.

Another consideration addressed in this work is how the train-test partitioning is performed. We hypothesize that, if

a parameter is highly redundant and provides limited information, as observed with V_{Bias} , an unbalanced pruning of this parameter, should not negatively impact performance. By unbalanced pruning, we mean a sampling strategy where the training set ensures proportional representation of all unique combinations of two of the parameters, while the number of unique values for the other parameter may vary or be underrepresented. To validate this theory, the dataset is split for training and testing in four different ways: randomly pruning all parameters, unbalanced pruning of V_{Bias} , unbalanced pruning of P_{data} , and unbalanced pruning of BW. This makes a total of 3 (models) $\times$ 9 (partitions) $\times$ 4 (pruning parameters) cases of study, as depicted in Fig. 3.

To determine the optimal hyperparameter configuration for each model, we used Optuna [15], an optimization tool that facilitates efficient hyperparameter tuning. Specifically, we employed Optuna's Tree-structured Parzen Estimator (TPE) sampler, a Bayesian optimization method that builds probabilistic models to identify promising regions in the hyperparameter space by focusing on areas likely to enhance objective function performance.

To evaluate the case studies, we began by establishing a baseline for the models using an 80% training and 20% testing split with random pruning across all parameters. Optimal hyperparameters were determined using Optuna with 5-fold cross-validation, and the resulting configurations are detailed in Table II. The models were then trained across all case variations, covering combinations of pruning conditions and different data partitions. Test set RMSE values for each FoM were extracted for every scenario, which are shown in Fig. 3.

TABLE II: Hyperparameter configurations for each model.

Model	Hyperparameter	Values
XGBoost	Number of estimators	776
	Maximum tree depth	5
GPR	Kernel type	Constant $\times$ RationalQuadratic
	Diagonal kernel bias, α	$8.0 \cdot 10^{-6}$
ANN	Number of hidden layers	3
	Neurons per layer	[2048, 512, 128]
	Activation	GELU

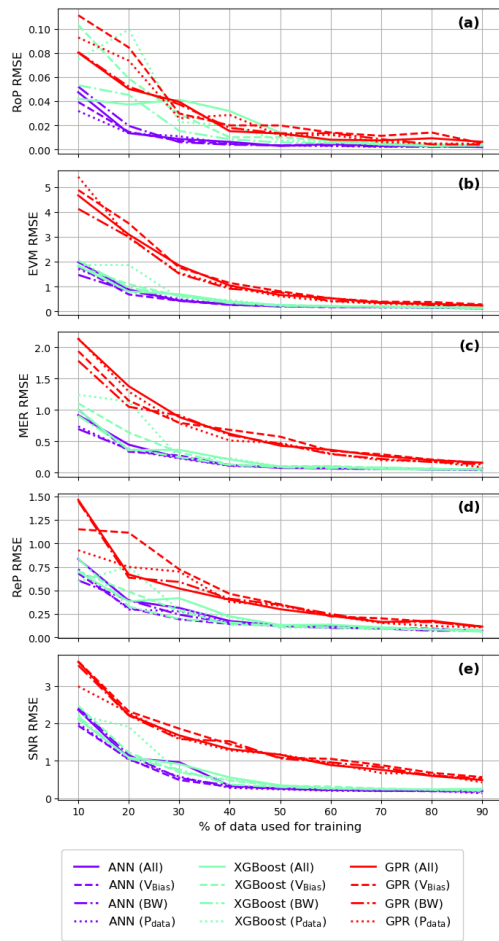


Fig. 3: Performance of the three ML models in terms of test RMSE with respect to the amount of data used for training. (a) RoP, (b) EVM, (c), MER, (d) ReP, (e) SNR.

V. RESULTS

In Figs. 3(a)-(e), the RMSE of RoP, EVM, MER, ReP and SNR metrics are depicted. Each figure presents the performance of the XGBoost model (green), GPR model (red), and ANN model (purple) across all four pruning cases.

First, it is notable that for a given model and data split, RMSE values show minimal variation regardless of the pruning strategy. This indicates that performance is influenced more by the overall amount of data available than by the specific representation of certain parameter values. However, deviations are observed in some cases, such as in Fig. 3(a) and (d), where the GPR model with a 20/80 train-test split and P_{data} unbalanced pruning performs worse than other pruning strategies. This discrepancy is likely due to statistical drift caused by the reduced data size, as the same RMSE trends are not observed with a 10/90 train-test split.

Then, comparing the models for the assessed FoMs, both the ANN and XGBoost demonstrate similar performance, with the ANN exhibiting a slight advantage. Both models significantly outperform the GPR, particularly when the training data is limited. This superior performance can be attributed to their

greater degrees of freedom, allowing them to approximate more complex functions, provided that sufficiently deep networks are employed. For instance, in Fig. 3(e), which shows the RMSE of the SNR, we observe that for a 50/50 train-test split with random pruning across all parameters, the ANN achieves a 77.3% improvement over GPR, while XGBoost shows a 70.1% improvement. With the exception of the RoP, where the performance between GPR and XGBoost is similar, the other metrics demonstrate comparable improvement ratios across the models for this case study.

Furthermore, GPR performance improves steadily with increasing data volume, though it fails to reach convergence in SNR. In contrast, XGBoost and ANN models also benefit from more data but achieve convergence much faster than GPR, with minimal to no further gains thereafter. For instance, considering the SNR RMSE in Fig. 3(e) for the ANN model with 50% data used for training, accurate results can already be obtained, indicating that the model has converged for this metric and cannot be improved further. This trend extends to the rest of metrics, and suggests that XGBoost and ANN may require significantly less data than GPR to reach optimal performance, which could be particularly beneficial in data-limited applications.

In general, the ANN would be preferable over the XGBoost model because it is compatible with gradient-based optimization techniques. Moreover, its training using 90% of the dataset reached convergence in 30 seconds on a 12th Gen Intel(R) Core(TM) i7-1260P CPU, with a memory footprint of only 4.29MB—highlighting its efficiency and suitability for real-world applications.

VI. PERFORMANCE-DRIVEN PARAMETER OPTIMIZATION WITH ANN-BASED DIGITAL TWIN

In this section, a proof-of-concept of an application of the proposed ANN-based digital twin is presented. More concretely, given a target FoM, an optimization algorithm finds the set of input parameters that provides such performance while minimizing the P_{data} using the DTON developed in previous sections. Specifically, we chose the ANN trained with 80% of the dataset with random pruning, which is the case of study with which the hyperparameters were optimized.

In this case, two global black-box optimization methods were employed: PSO with a swarm size of 100, inertia weight of 0.5, and cognitive/social coefficients of 0.5, and differential evolution (DE) with a population size of 15, mutation factor of 0.8, and crossover rate of 0.7, both yielding similar results and taking less than 2 seconds.

The process to evaluate their performance is the following: first, a sample is selected from the test set with a certain target set of FoMs. Then, the optimization algorithm iterates utilizing the pretrained ANN until it finds the combination of input parameters that provides the closest QoT metrics. The P_{data} is included as a penalty, forcing the solution with minimum power consumption. Afterwards, the RMSE between the predicted and target set of FoMs is computed and, finally, the P_{data} difference between the parameters utilized in the

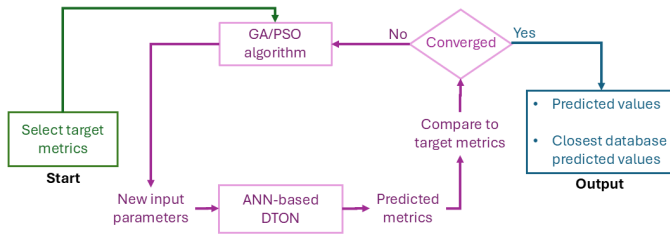


Fig. 4: Flow diagram to obtain the optimal parameters for a target set of FoMs.

		RMSE					Avg ΔP_{data} (dB)
		RoP	EVM	MER	ReP	SNR	
Predicted FoM	GA	0.000	0.397	0.187	0.202	0.167	0.552 dB
	PSO	0.000	0.412	0.234	0.200	0.198	0.348 dB
Closest true FoM	GA	0.000	0.978	0.312	0.412	0.851	0.552 dB
	PSO	0.000	0.966	0.366	0.506	0.653	0.351 dB

TABLE III: RMSE comparison of predicted FoM and average power reduction between DE and PSO on the test set with an ANN-based DTON.

dataset and the ones obtained by the algorithm is calculated. Moreover, the optimized parameters are mapped to the closest actual values in our database, and the resulting true metrics are compared to the target metrics. This ensures that ANN predictions are realistic, as low errors indicate that points in the database near the predictions align closely with them. Fig. 4 depicts the described procedure. This process is repeated all over the test set, and the average results are provided in Table III.

Both optimization algorithms are capable of successfully identifying parameter configurations that yield low error, with an RMSE below 1.0 in every metric. This indicates that a reliable control loop for the optical system could be implemented using either optimization algorithm with the proposed DTON. Additionally, the average power reduction shown in the table, demonstrates that P_{data} can be reduced in our case by up to 0.552 dB when comparing the optimized parameter configurations to suboptimal dataset solutions for a given target metric, highlighting the system’s effectiveness in resource optimization.

VII. CONCLUSIONS

In this work, a machine learning-based digital twin for predicting QoT metrics in a photonic mmW signal generation fronthaul link has been developed. Using a simulation model previously validated with experimental results, a dataset of 1100 samples was generated, where each data point maps a specific configuration of parameters (P_{data} , BW, and V_{Bias}) to its respective FoM—RoP, EVM, MER, ReP, and SNR—at the receiver. After curating the dataset to optimize model performance, three different ML models were evaluated: a GPR model, an ANN, and an XGBoost model. Comparisons based on RMSE across all five tested FoMs revealed that the XGBoost and ANN models performed similarly, both outperforming the GPR model.

Thus, the proposed ML-based DTON bridges the gap between physics-based models and machine learning ap-

proaches through a hybrid data-driven method, enabling real-time network optimization without direct system intervention. By reducing reliance on extensive hardware monitoring, the approach enhances cost-effectiveness and scalability. Additionally, it integrates optimization algorithms such as PSO and GA to fine-tune network configurations—an aspect not commonly combined with real-time Quality of Transmission (QoT) estimation.

Future work will focus on transfer learning between different real world optical setups, utilizing physics informed neural networks (PINNs) for mirror modelling, and exploring alternative optimization algorithms to find the optimum setup configurations.

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